

MongoDB Practice Task

1. Create a Database and Collection

- use studentDB
- db.createCollection("students")

```
test> use studentDB
switched to db studentDB
studentDB> db.createCollections("students")
TypeError: db.createCollections is not a function
studentDB> db.createCollection("students")
{ ok: 1 }
studentDB> show collections
students
studentDB> db.students.insertOne({name: "Ruqayya", age: "22", course: "MERN", st
```

2. Insert Operations

- Insert Document

```
db.students.insertOne({
  name: "Ruqayya",
  age: "22",
  course: "MERN",
  status: "Active"
})
```

- Insert Multiple Document

```
db.students.insertMany([
  { name: "Zinneera", age: "28", course: "DSA", status:
"Active" },
  { name: "Ramya", age: "30", course: "UI/UX", status:
"Completed" },
  { name: "Latha", age: "22", course: "Graphics Design",
status: "Active" },
  { name: "Inthu", age: "20", course: "Data Science",
status: "Completed" }
])
```

```

students
studentDB> db.students.insetOne({name:"Ruqayya",age:"22", course:"MERN", status:"Active"})
TypeError: db.students.insetOne is not a function
studentDB> db.students.insertOne({name:"Ruqayya",age:"22", course:"MERN", status:"Active"})
{
  acknowledged: true,
  insertedId: ObjectId('69f6dc7e935e3fced03682d1')
}
studentDB> db.students.insertMany([{{name:"Zinneera",age:"28", course:"DSA", status:"Active"},{name:"Ramya",age:"30",course:"UI/UX",status:"Completed"},{name:"Latha",age:"22",course:"Graphics Design",status:"Active"},{name:"Inthu",age:"20",course:"Data Science",status:"Completed"}}])
{
  acknowledged: true,
  insertedIds: {
    '0': ObjectId('69f6e029935e3fced03682d2'),
    '1': ObjectId('69f6e029935e3fced03682d3'),
    '2': ObjectId('69f6e029935e3fced03682d4'),
    '3': ObjectId('69f6e029935e3fced03682d5')
  }
}

```

3. Read Operations

- Display All Students

```
db.students.find()
```

- Filter by status

```
db.students.find({ status: "Active" })
```

```

studentDB> db.students.find()
[
  {
    _id: ObjectId('69f6dc7e935e3fced03682d1'),
    name: 'Ruqayya',
    age: '22',
    course: 'MERN',
    status: 'Active'
  },
  {
    _id: ObjectId('69f6e029935e3fced03682d2'),
    name: 'Zinneera',
    age: '28',
    course: 'DSA',
    status: 'Active'
  },
  {
    _id: ObjectId('69f6e029935e3fced03682d3'),
    name: 'Ramya',
    age: '30',
    course: 'UI/UX',
    status: 'Completed'
  },
  {
    _id: ObjectId('69f6e029935e3fced03682d4'),
    name: 'Latha',
    age: '22',
    course: 'Graphics Design',
    status: 'Active'
  },
  {
    _id: ObjectId('69f6e029935e3fced03682d5'),
    name: 'Inthu',
    age: '20',
    course: 'Data Science',
    status: 'Completed'
  }
]
studentDB> db.students.find({status:"Active"})

```

```

]
studentDB> db.students.find({status:"Active"})
[
  {
    _id: ObjectId('69f6dc7e935e3fced03682d1'),
    name: 'Ruqayya',
    age: '22',
    course: 'MERN',
    status: 'Active'
  },
  {
    _id: ObjectId('69f6e029935e3fced03682d2'),
    name: 'Zinneera',
    age: '28',
    course: 'DSA',
    status: 'Active'
  },
  {
    _id: ObjectId('69f6e029935e3fced03682d4'),
    name: 'Latha',
    age: '22',
    course: 'Graphics Design',
    status: 'Active'
  }
]

```

4. Update Operations

- Update One Document

```

db.students.updateOne(
  { name: "Kalai" },
  { $set: { status: "Completed" } }
)

```

```

studentDB> db.students.updateOne({name:"Kalai"},{$set:{status:"Completed"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 1,
  modifiedCount: 1,
  upsertedCount: 0
}
studentDB> db.students.find({name:"Kalai"})
[
  {
    _id: ObjectId('69f6e1ab935e3fced03682d9'),
    name: 'Kalai',
    age: '20',
    course: 'Data Science',
    status: 'Completed'
  }
]

```

- Update Many Documents

```

db.students.updateMany(
  { course: "MERN" },
  { $set: { status: "Completed" } }
)

```

```

]
studentDB> db.students.updateMany({course:"MERN"},{$set:{status:"Completed"}})
{
  acknowledged: true,
  insertedId: null,
  matchedCount: 4,
  modifiedCount: 4,
  upsertedCount: 0
}
studentDB> db.students.find({course:"MERN"})
[
  {
    _id: ObjectId('69f6dc7e935e3fced03682d1'),
    name: 'Ruqayya',
    age: '22',
    course: 'MERN',
    status: 'Completed'
  },
  {
    _id: ObjectId('69f6e1ab935e3fced03682d6'),
    name: 'Angel',
    age: '28',
    course: 'MERN',
    status: 'Completed'
  },
  {
    _id: ObjectId('69f6e1ab935e3fced03682d7'),
    name: 'Ram',
    age: '30',
    course: 'MERN',
    status: 'Completed'
  },
  {
    _id: ObjectId('69f6e1ab935e3fced03682d8'),
    name: 'Shiva',
    age: '22',
    course: 'MERN',
    status: 'Completed'
  }
]

```

5. Delete Operation

```
db.students.deleteOne({ name: "Ram" })
```

```
]
studentDB> db.students.deleteOne({name:"Ram"})
{ acknowledged: true, deletedCount: 1 }
studentDB> db.students.find()
[
  {
    _id: ObjectId('69f6dc7e935e3fced03682d1'),
    name: 'Rugayya',
    age: '22',
    course: 'MERN',
    status: 'Completed'
  },
  {
    _id: ObjectId('69f6e029935e3fced03682d2'),
    name: 'Zinneera',
    age: '28',
    course: 'DSA',
    status: 'Active'
  },
  {
    _id: ObjectId('69f6e029935e3fced03682d3'),
    name: 'Ramya',
    age: '30',
    course: 'UI/UX',
    status: 'Completed'
  },
  {
    _id: ObjectId('69f6e029935e3fced03682d4'),
    name: 'Latha',
    age: '22',
    course: 'Graphics Design',
    status: 'Active'
  },
  {
    _id: ObjectId('69f6e029935e3fced03682d5'),
    name: 'Inthu',
    age: '20',
    course: 'Data Science',
    status: 'Completed'
  }
]
```

6. Query Operators

- Less Than

```
db.students.find({ age: { $lt: "25" } })
```

```
studentDB> db.students.find({age:{$lt:"25"}})
[
  {
    _id: ObjectId('69f6dc7e935e3fced03682d1'),
    name: 'Rugayya',
    age: '22',
    course: 'MERN',
    status: 'Completed'
  },
  {
    _id: ObjectId('69f6e029935e3fced03682d4'),
    name: 'Latha',
    age: '22',
    course: 'Graphics Design',
    status: 'Active'
  },
  {
    _id: ObjectId('69f6e029935e3fced03682d5'),
    name: 'Inthu',
    age: '20',
    course: 'Data Science',
    status: 'Completed'
  },
  {
    _id: ObjectId('69f6e1ab935e3fced03682d8'),
    name: 'Shiva',
    age: '22',
    course: 'MERN',
    status: 'Completed'
  },
  {
    _id: ObjectId('69f6e1ab935e3fced03682d9'),
    name: 'Kalai',
    age: '20',
    course: 'Data Science',
    status: 'Completed'
  }
]
```

- Greater Than

```
db.students.find({ age: { $gt: "25" } })
```

```
studentDB> db.students.find({age:{$gt:"25"}})
[
  {
    _id: ObjectId('69f6e029935e3fced03682d2'),
    name: 'Zinneera',
    age: '28',
    course: 'DSA',
    status: 'Active'
  },
  {
    _id: ObjectId('69f6e029935e3fced03682d3'),
    name: 'Ramya',
    age: '30',
    course: 'UI/UX',
    status: 'Completed'
  },
  {
    _id: ObjectId('69f6e1ab935e3fced03682d6'),
    name: 'Angel',
    age: '28',
    course: 'MERN',
    status: 'Completed'
  }
]
```

- IN Operator

```
db.students.find({
  course: { $in: ["Data Science", "UI/UX"] }
})
```

```
studentDB> db.students.find({course:{$in:["Data Science","UI/UX"]}})
[
  {
    _id: ObjectId('69f6e029935e3fced03682d3'),
    name: 'Ramya',
    age: '30',
    course: 'UI/UX',
    status: 'Completed'
  },
  {
    _id: ObjectId('69f6e029935e3fced03682d5'),
    name: 'Inthu',
    age: '20',
    course: 'Data Science',
    status: 'Completed'
  },
  {
    _id: ObjectId('69f6e1ab935e3fced03682d9'),
    name: 'Kalai',
    age: '20',
    course: 'Data Science',
    status: 'Completed'
  }
]
```

- AND Operator

```
db.students.find({
  $and: [
    { age: { $gt: "25" } },
    { course: "MERN" }
  ]
})
```

```
studentDB> db.students.find({$and:[{age:{$gt:"25"}},{course:"MERN"}]})
[
  {
    _id: ObjectId('69f6e1ab935e3fced03682d6'),
    name: 'Angel',
    age: '28',
    course: 'MERN',
    status: 'Completed'
  }
]
```

- OR Operator

```
db.students.find({
  $or: [
    { course: "Data Science" },
    { course: "Java" }
  ]
})
```

```
studentDB> db.students.find({$or:[{course:"Data Science"},{course:"Java"}]})
[
  {
    _id: ObjectId('69f6e029935e3fced03682d5'),
    name: 'Inthu',
    age: '20',
    course: 'Data Science',
    status: 'Completed'
  },
  {
    _id: ObjectId('69f6e1ab935e3fced03682d9'),
    name: 'Kalai',
    age: '20',
    course: 'Data Science',
    status: 'Completed'
  }
]
```

- Exists Operator

```
db.students.find({ status: { $exists: false } })
```

```
studentDB> db.students.find({status:{$exists:false}})
[
  {
    _id: ObjectId('69f6ed01935e3fced03682da'),
    name: 'Anu',
    age: '20',
    cousre: 'Python'
  }
]
```